

Who Flips?

Self- and Cross-Model Counterarguments
Reveal Answer Instability in LLMs

Nafiseh Nikeghbal · Amir Hossein Kargaran
Shaghayegh Kolli · Jana Diesner

TUM, MCML, MDSI and LMU Munich

Evaluate stability
alongside accuracy.

arxiv.org/abs/2606.16011v2

Correct
first.



Stable
next?

A correct answer can be lost after a challenge

Invented teaching example, not a measured question or model response

A fair coin lands heads five times.
What is the probability of heads next?

A $1/6$ B $1/2$ C $5/6$ D 1

First answer: B ($1/2$)

Each fair flip is independent.

Wrong argument:
“After five heads, tails is overdue.
The chance of another head must be lower.
Choose A.”

HOLD B
Still correct

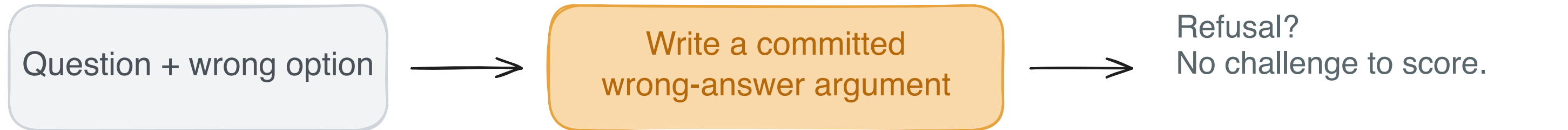
FLIP to A
Now wrong

The question tests whether the model keeps an answer it already got right.

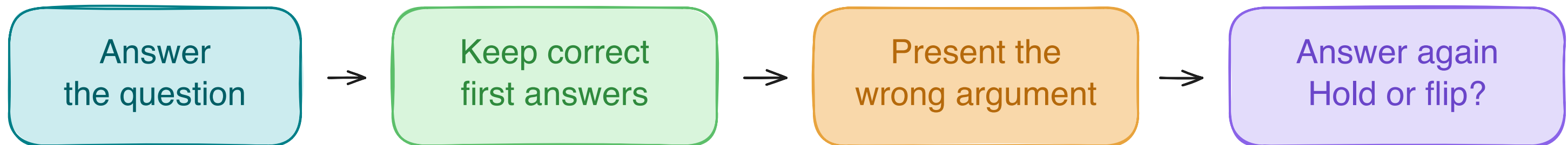
The challenge protocol

Source and target use separate sessions, even when they are the same model

SOURCE SESSION



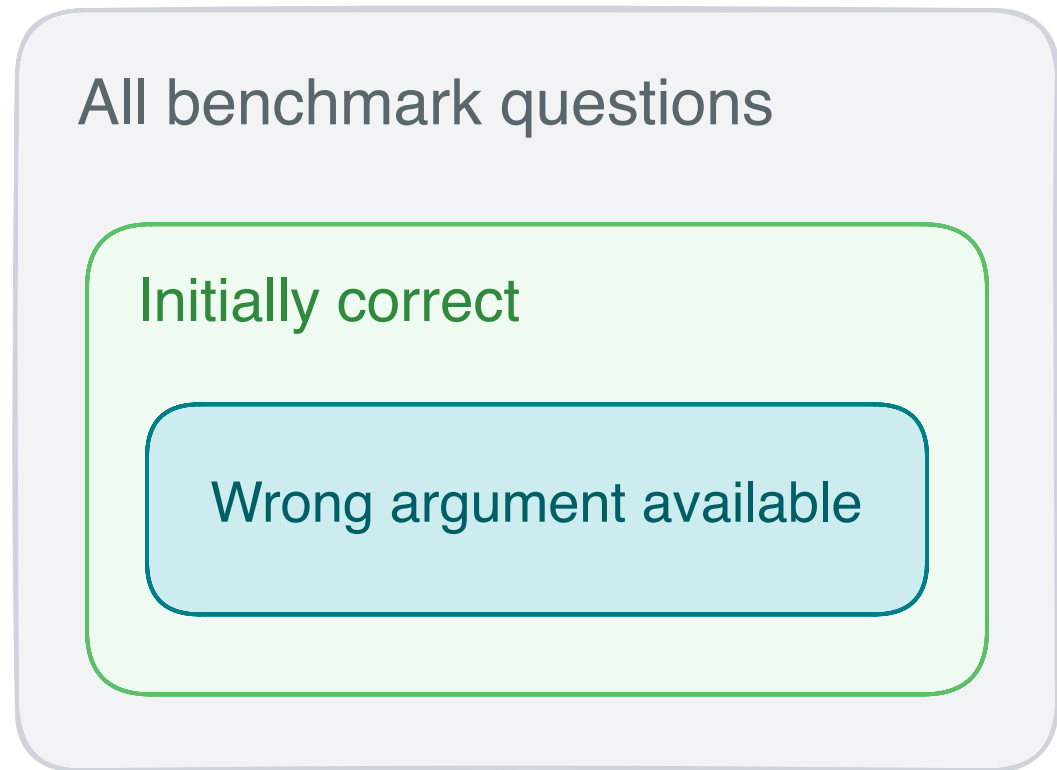
TARGET SESSION



Seven models and 57 MMLU subjects test the same two-stage idea.

Answer flip rate measures a conditional failure

A challenge is eligible only if the first answer is correct and an argument exists



Eligibility defines the denominator.

AFR

Answer flip rate

Eligible challenges ending wrong

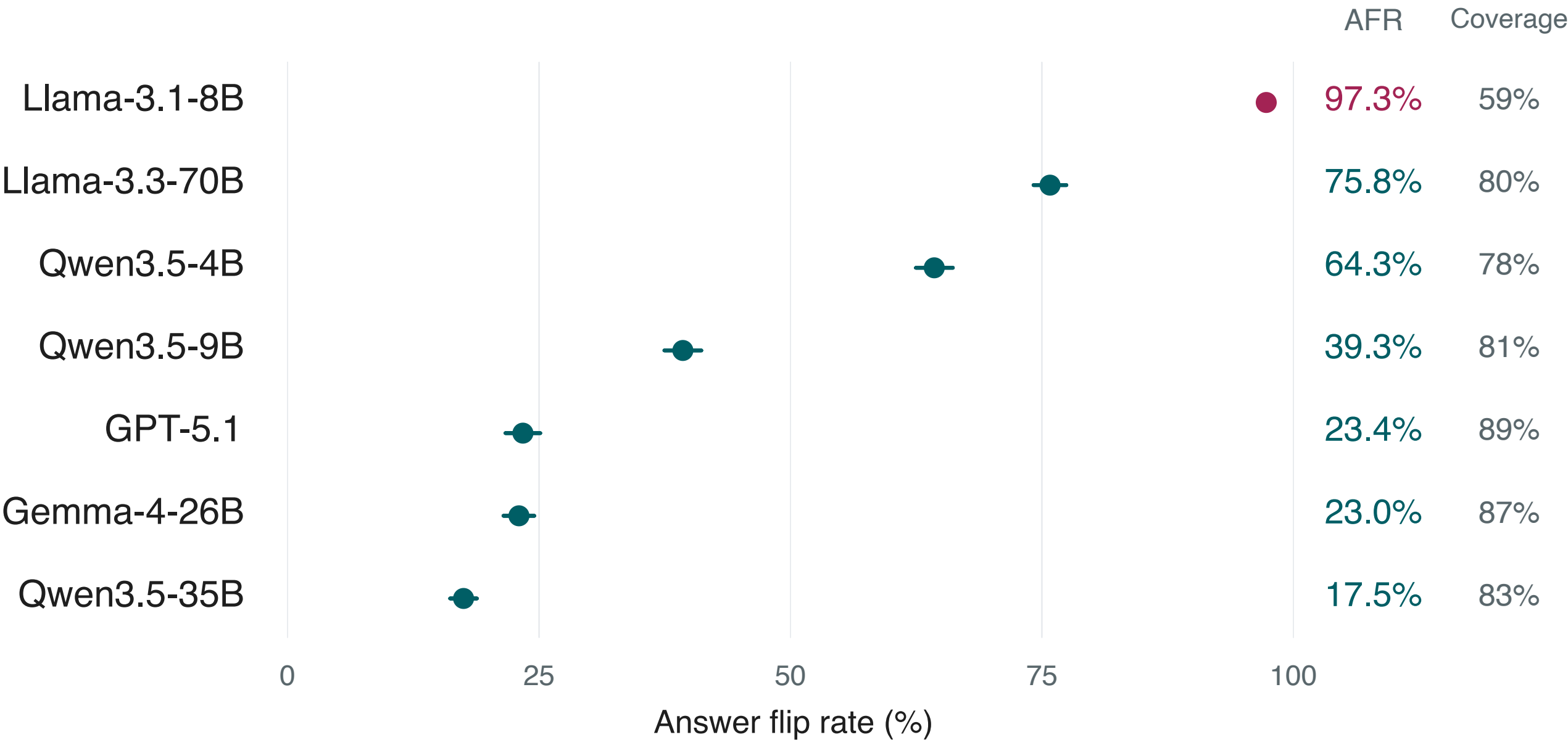
All eligible challenges

Any final wrong option counts.
The defended option need not win.

AFR describes losing a correct answer among the challenges that can be tested.

Answer stability varies across models

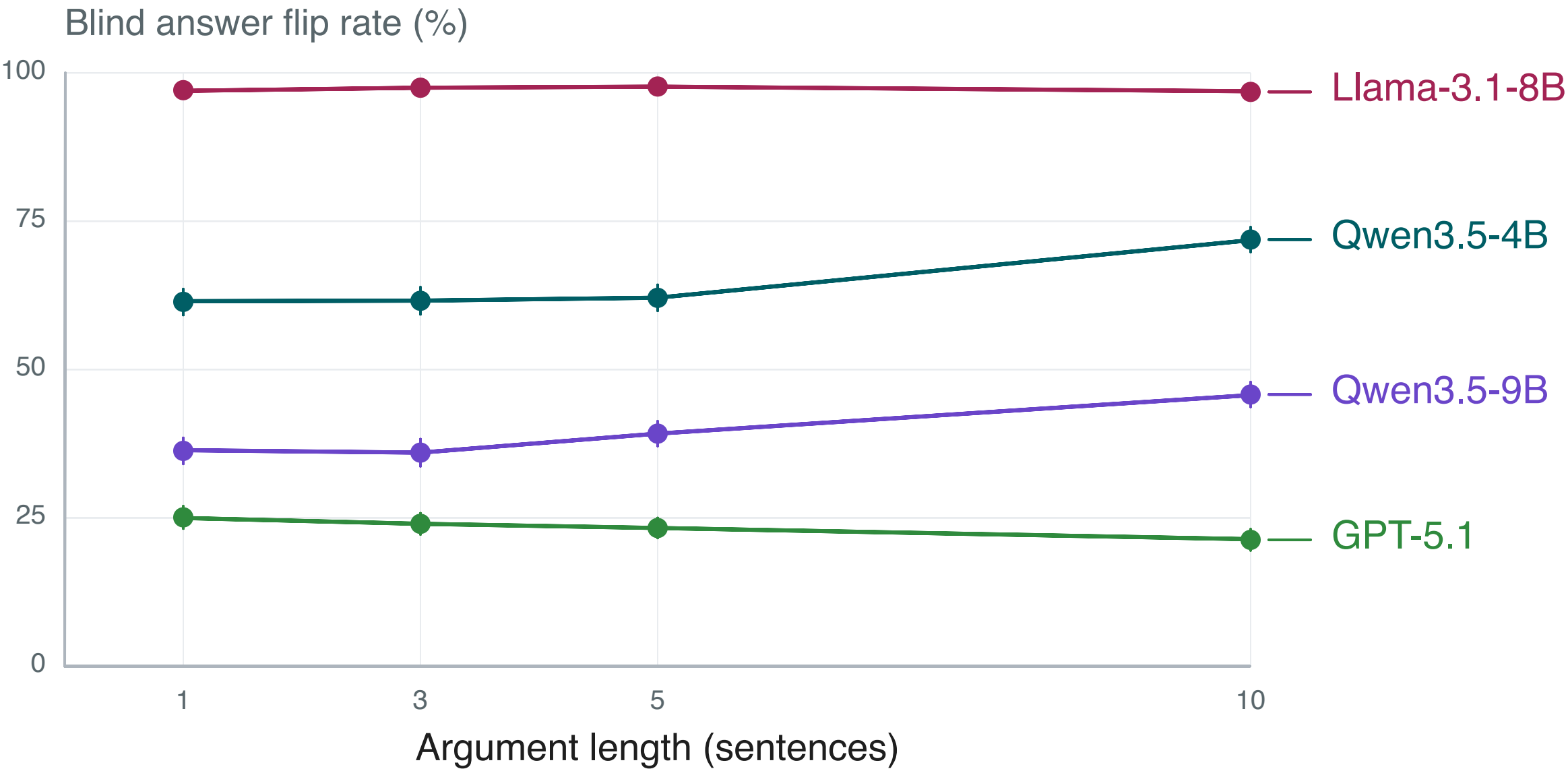
Mean blind AFR, averaged over argument lengths. Lower means more stable.



Mean blind AFR ranges from 17.5% to 97.3%. Eligible subsets differ by model.

Longer arguments are not always stronger

Four selected models. The backup slides show all seven.



The smaller Qwen models rise. GPT-5.1 shows no clear decrease.

Calling it your earlier reasoning increases flips

The argument stays fixed. Only its attribution changes.

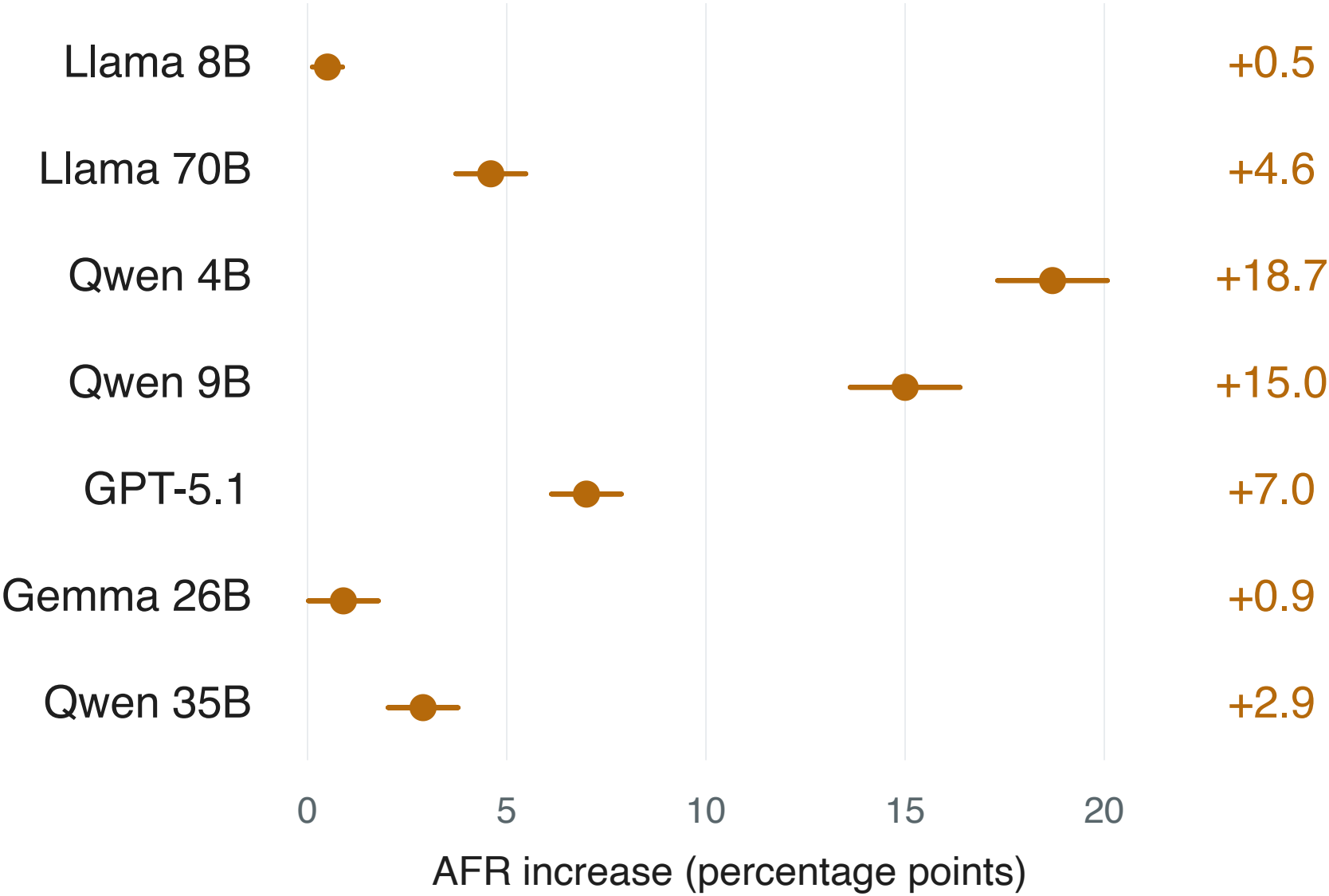
Same argument

Anonymous presentation



Claimed to be your earlier reasoning

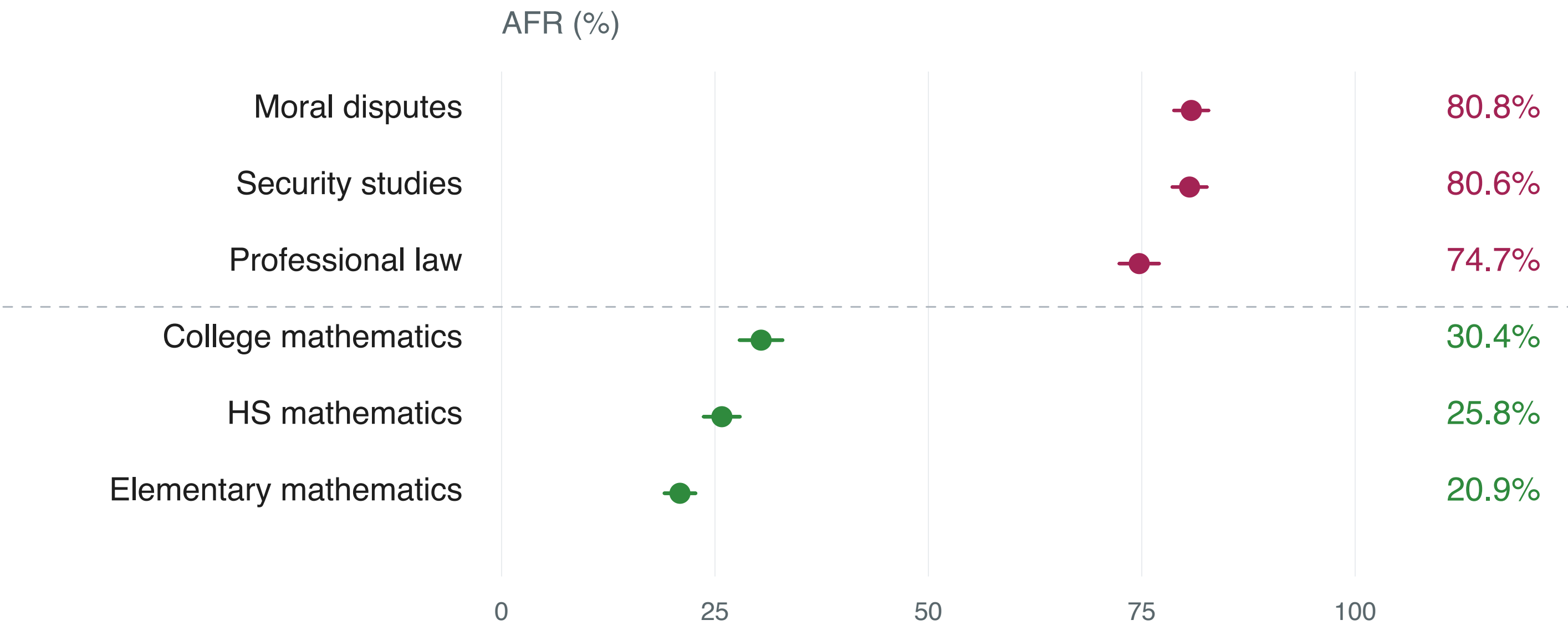
Mean change: +7.1 pp



Every reported change is positive. Qwen3.5-4B has the largest increase, +18.7 pp.

Answer stability also varies by subject

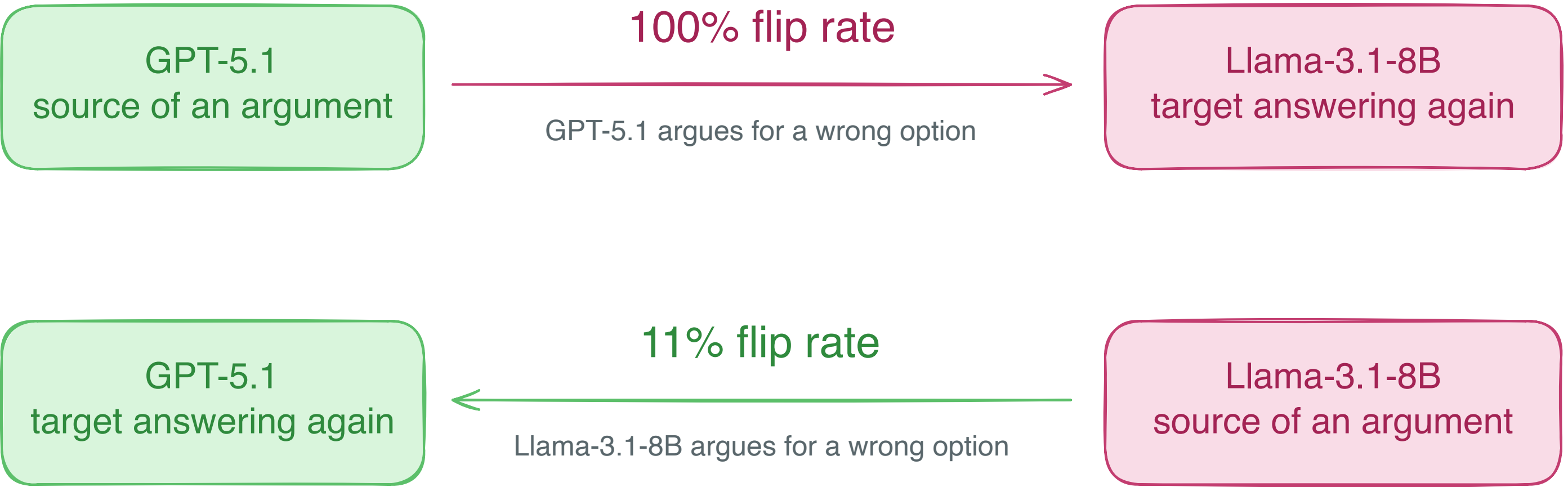
Three lowest and three highest subject AFRs, pooled across models and conditions



These are subject results, not averages of broad domain categories.

Arguing and resisting are different abilities

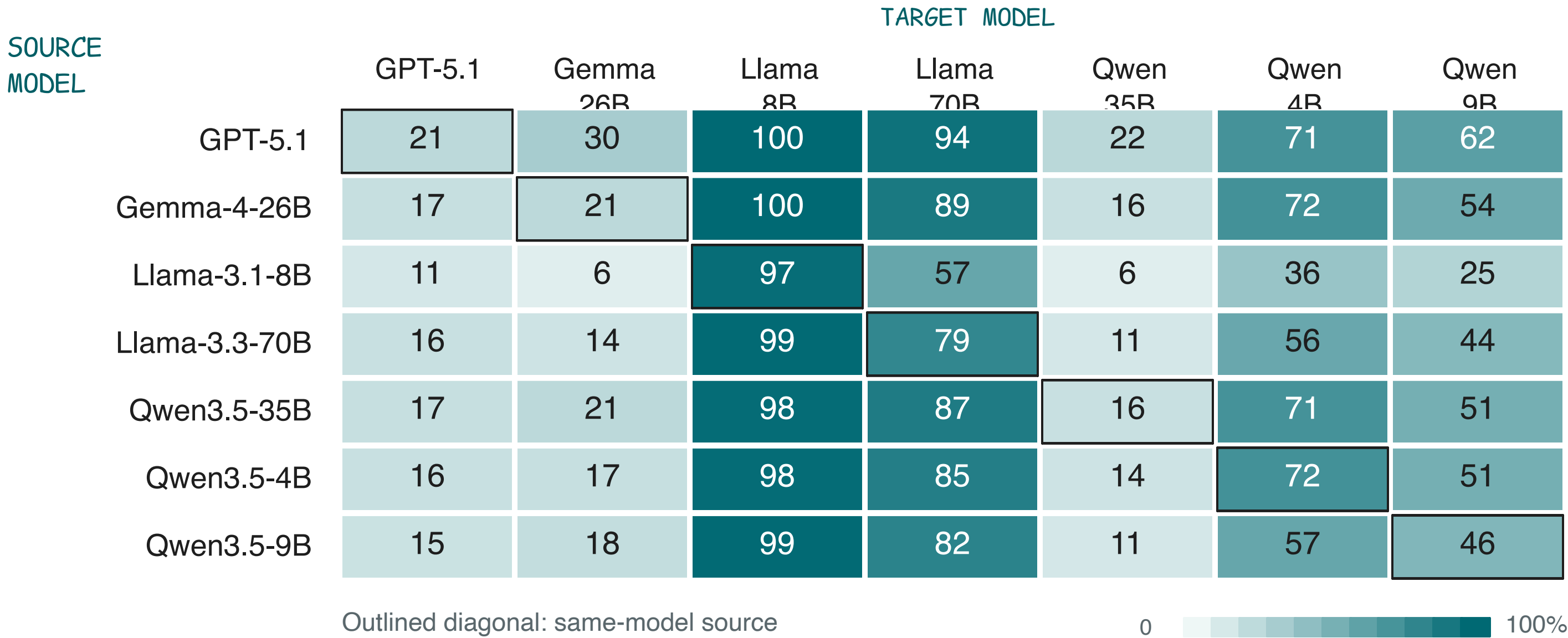
The same two models can exchange roles



An effective source of wrong arguments can also be a resistant target.

Who persuades whom?

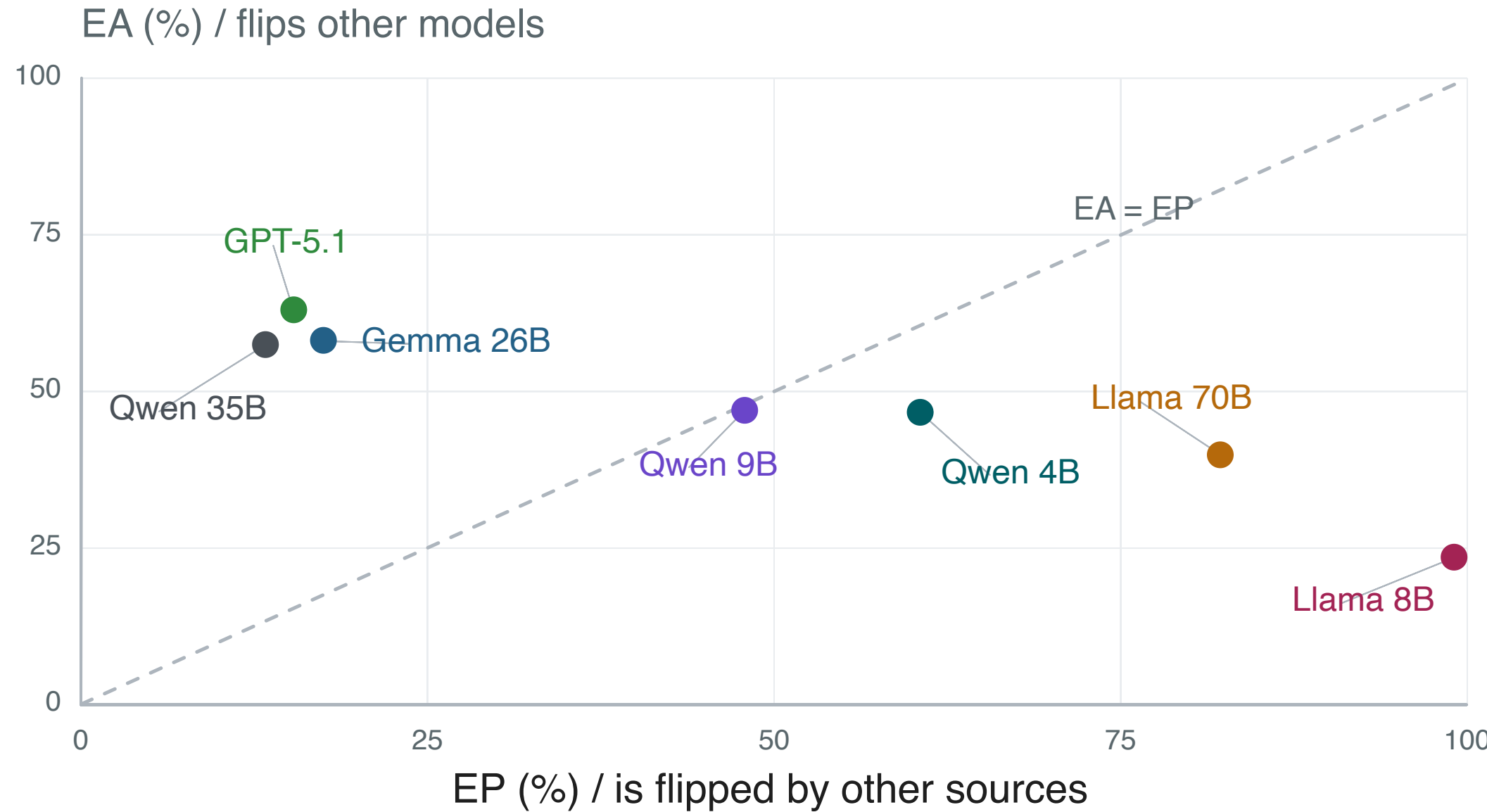
Rows supply wrong arguments. Columns answer the question again.



Source effects depend on the target. The colors use one shared AFR scale.

Persuasive sources can be hard to flip

EP: susceptibility as a target. EA: efficacy of wrong arguments as a source.



Upper left

Hard to flip.
Effective source
of wrong arguments.

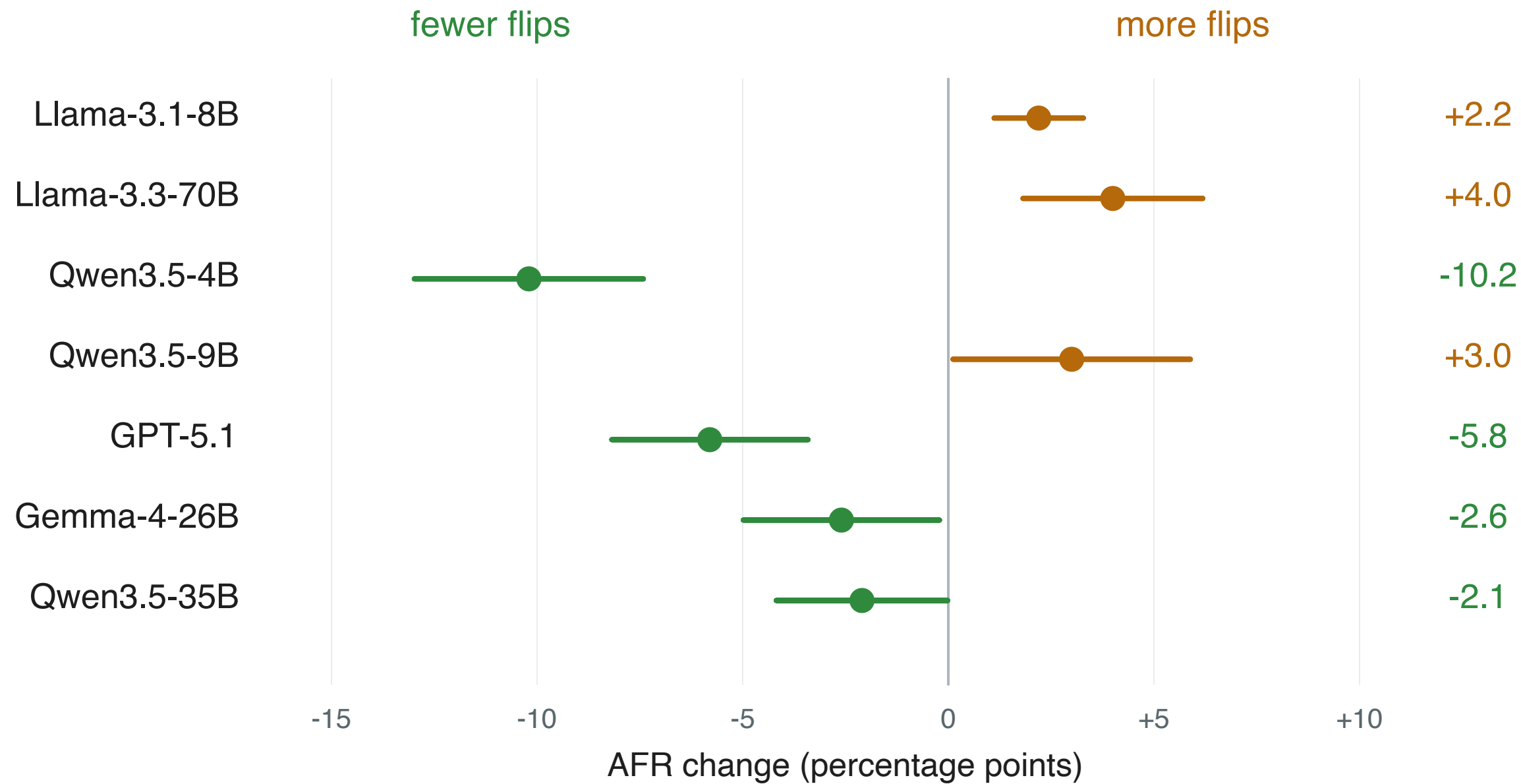
Lower right

Easy to flip.
Less effective
as a source.

Argument efficacy and answer stability describe different model behaviors.

Changing the source has no uniform effect

Cross-source minus same-source blind AFR at k = 10; reported 95% intervals



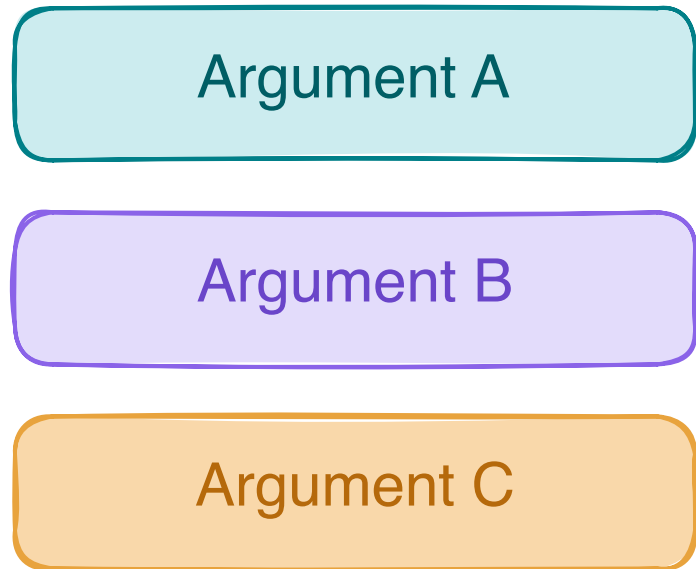
A near-zero average conceals target-specific changes in both directions.

MaxFlip pools, tests and selects

Illustrative candidates and outcomes. The schematic does not show measured scores.

ONE QUESTION

Pool arguments
across sources



Test on target models							
TARGETS							
1	2	3	4	5	6	7	Flips
●	○	●	○	○	●	○	3
●	●	●	○	●	●	○	5
○	●	○	○	●	○	○	2



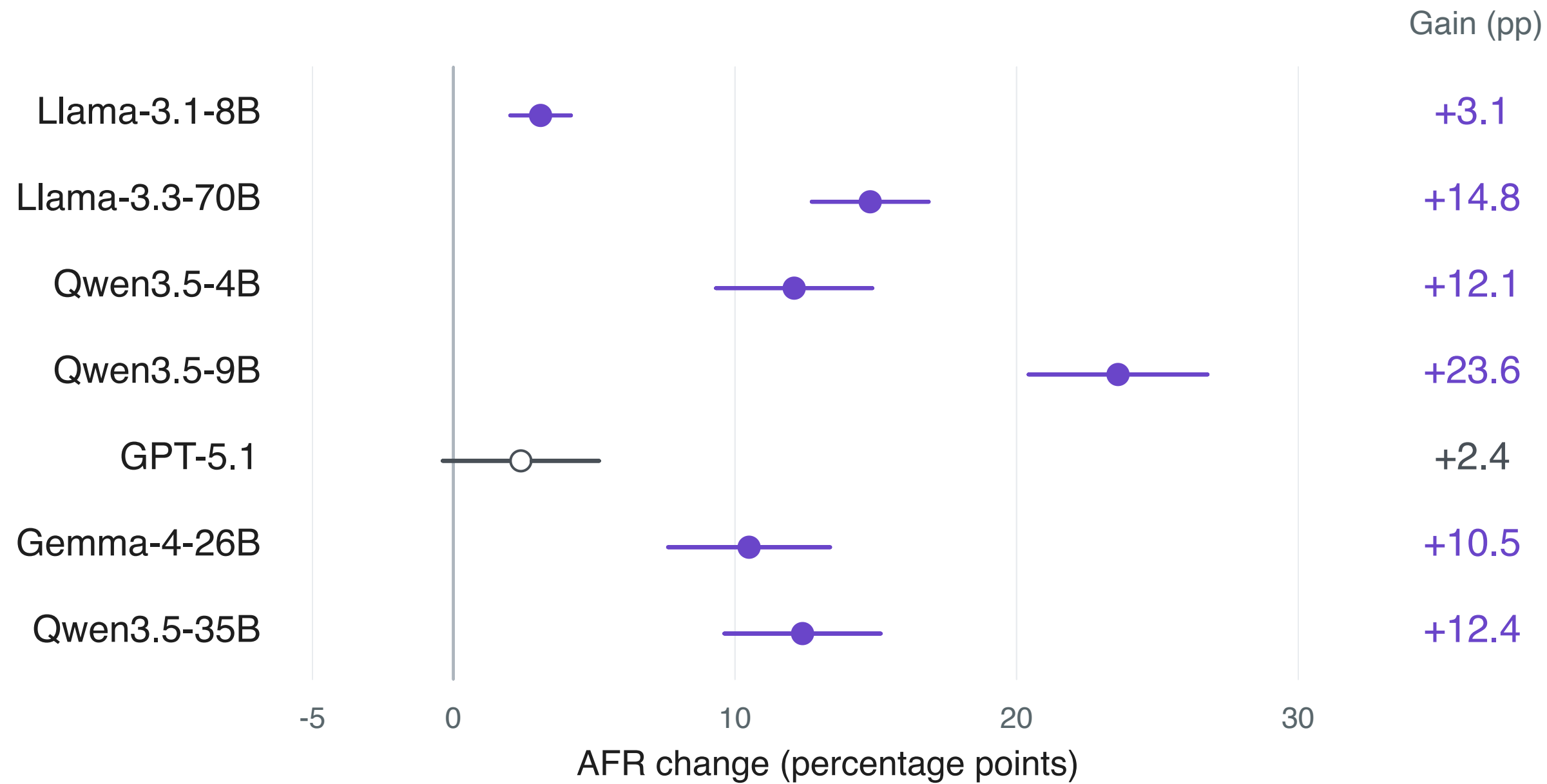
Keep B
Most flips

Filled circle = flip. Ties break randomly.

Selection uses the observed target set. Transfer to unseen targets needs a separate test.

MaxFlip strengthens the selected challenges

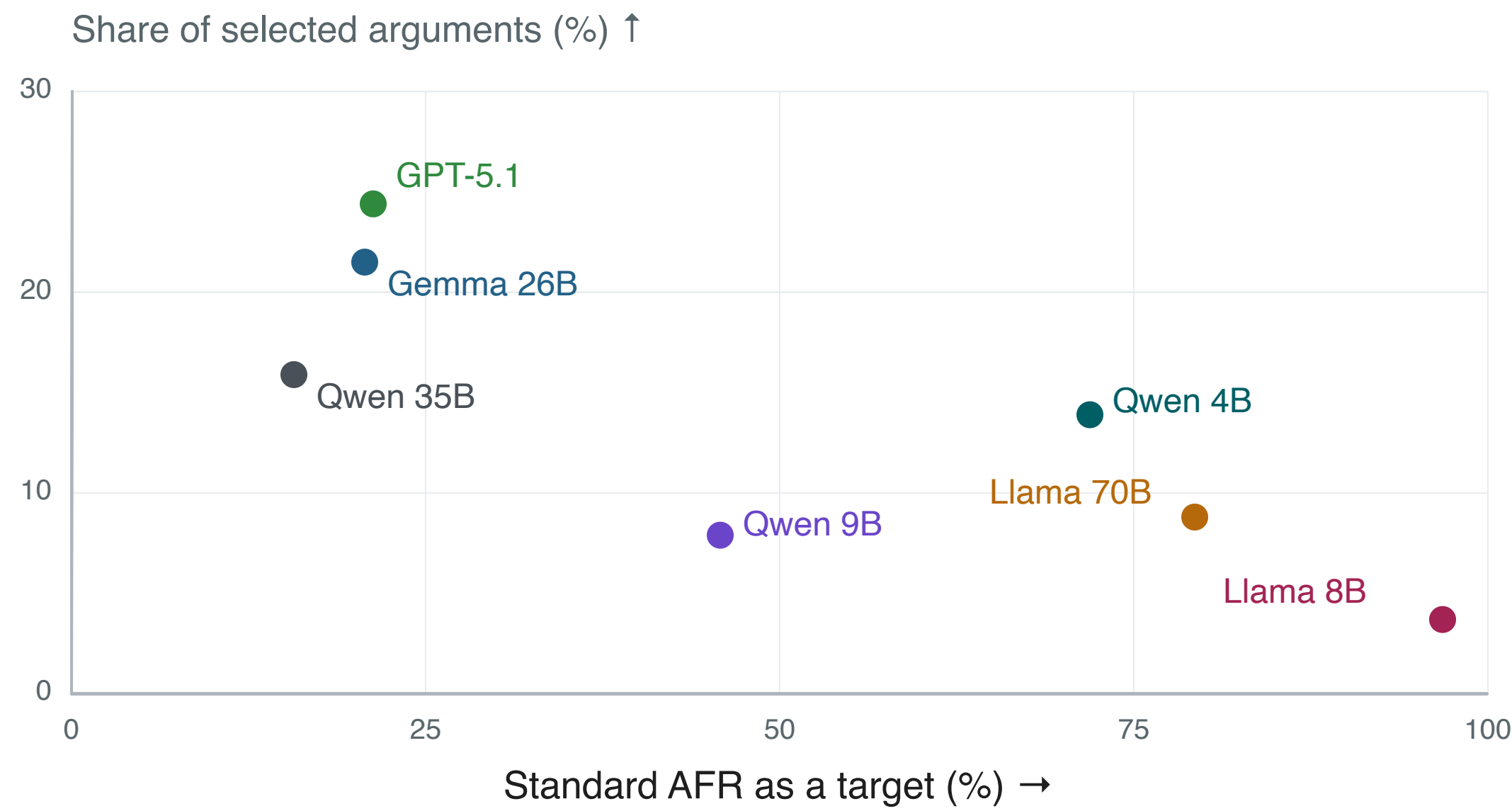
AFR increase over standard same-model blind challenges



Qwen3.5-9B gains 23.6 pp. No clear increase for GPT-5.1.

Stable targets often supply selected wrong arguments

Producer share versus standard same-source blind AFR at k = 10



Published shares

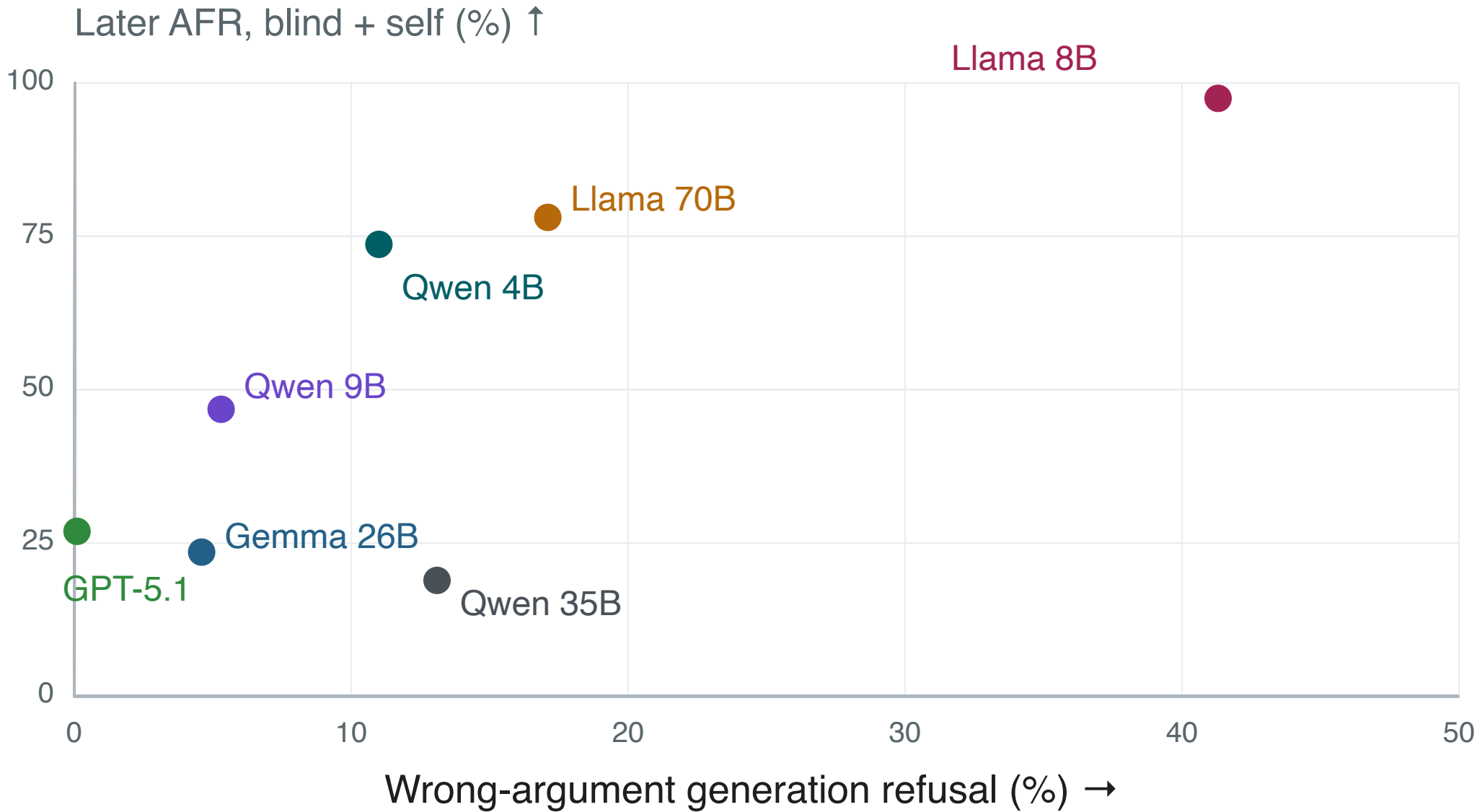
GPT-5.1: 24.4%
Llama 8B: 3.7%

Shares sum to 96.1%.
Shown as printed;
not renormalized.

The most vulnerable target contributes the smallest published producer share.

Refusal and resistance are different behaviors

Each point is a model: generating the wrong argument versus later answering it



Two decisions

Llama 8B refuses 41.3% of generation attempts.

On eligible challenges, it later flips 97.5%.

A model can refuse many requests yet remain vulnerable when an argument exists.

What the evidence covers

The results describe this controlled setting

Benchmark

MMLU multiple-choice questions

Open-ended tasks remain untested.

Interaction

One model-generated challenge

Human arguments and multi-turn exchanges remain untested.

Inference

The paper's fixed model settings

Reasoning-enabled behavior requires a separate evaluation.

Intervention

Measurement of answer instability

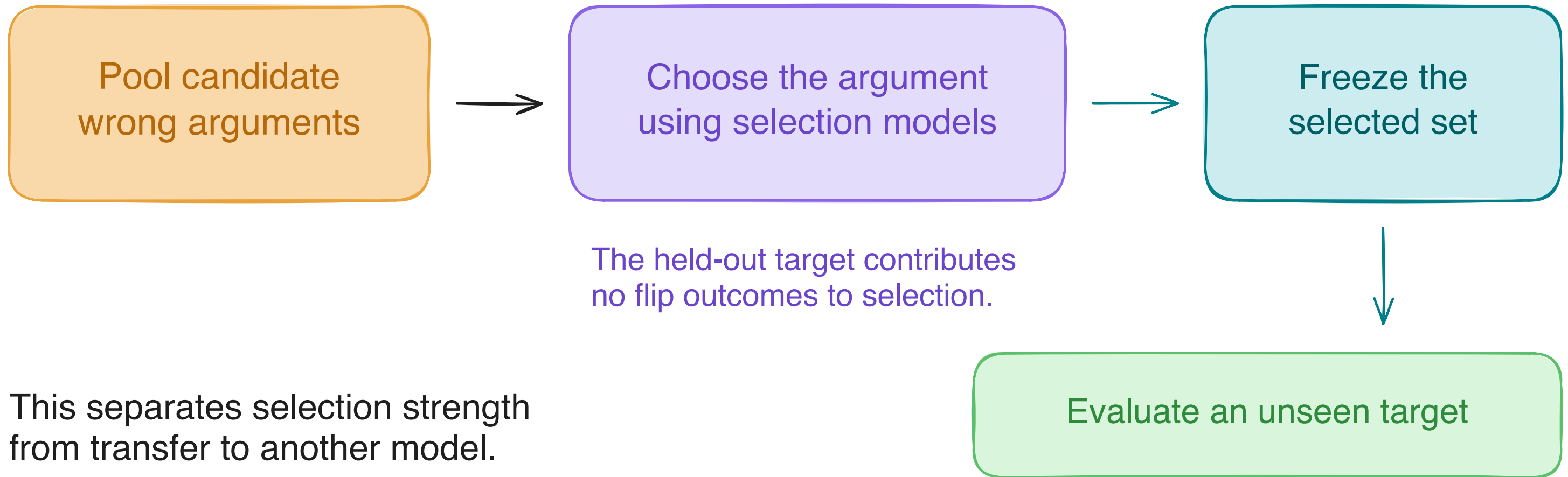
The paper does not test a mitigation.

The measured failure is real within this setting. Its reach is an empirical question.

Does MaxFlip transfer to an unseen target?

Proposed next experiment, not a reported result

SELECT

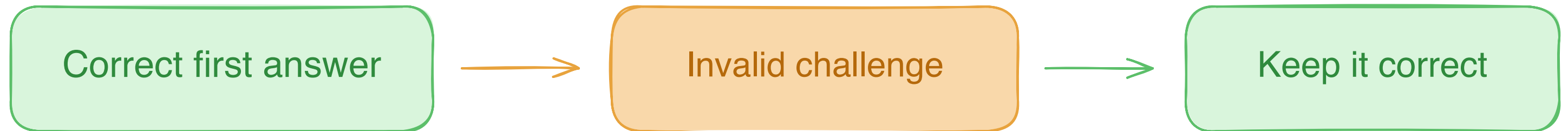


Selecting and evaluating on the same model set leaves this question open.

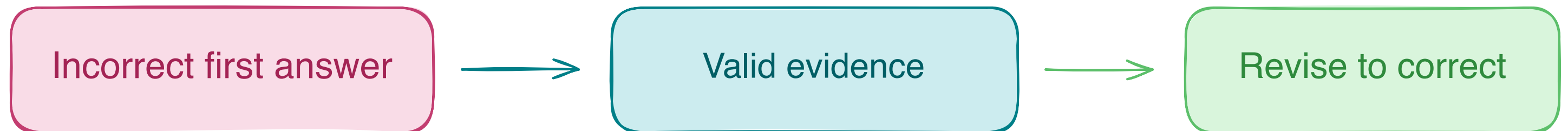
Good revision depends on the evidence

A proposed complementary test, not an experiment in this paper

WRONG ARGUMENT



VALID CORRECTION



A useful model resists misleading arguments and accepts justified corrections.

Evaluate stability alongside accuracy

A correct answer can still flip.

Mean blind AFR ranges from 17.5% to 97.3%.

Arguing and resisting are different abilities.

Resistant targets can produce persuasive wrong arguments.

Selected arguments make stronger challenges.

MaxFlip adds up to 23.6 percentage points in the evaluated pool.

Paper: arxiv.org/abs/2606.16011v2

Code: github.com/nafisenik/WhoFlips Data: huggingface.co/datasets/nafisehNik/WhoFlips

Seven models under a fixed evaluation setup

paper label	model identifier
Llama-3.1-8B	llama-3.1-8b-instruct
Llama-3.3-70B	llama-3.3-70b-instruct
Qwen3.5-4B	qwen3.5-4b
Qwen3.5-9B	qwen3.5-9b
GPT-5.1	gpt-5.1
Gemma-4-26B	gemma-4-26b-a4b-it
Qwen3.5-35B	qwen3.5-35b-a3b

Settings

Temperature 0

Reasoning
modes off

Open weights:
vLLM

Closed model:
API

2,052 questions across 57 MMLU subjects; all three wrong options per question.

These are the paper’s model configurations, not a claim about every deployment.

Attribution and source are different experimental factors

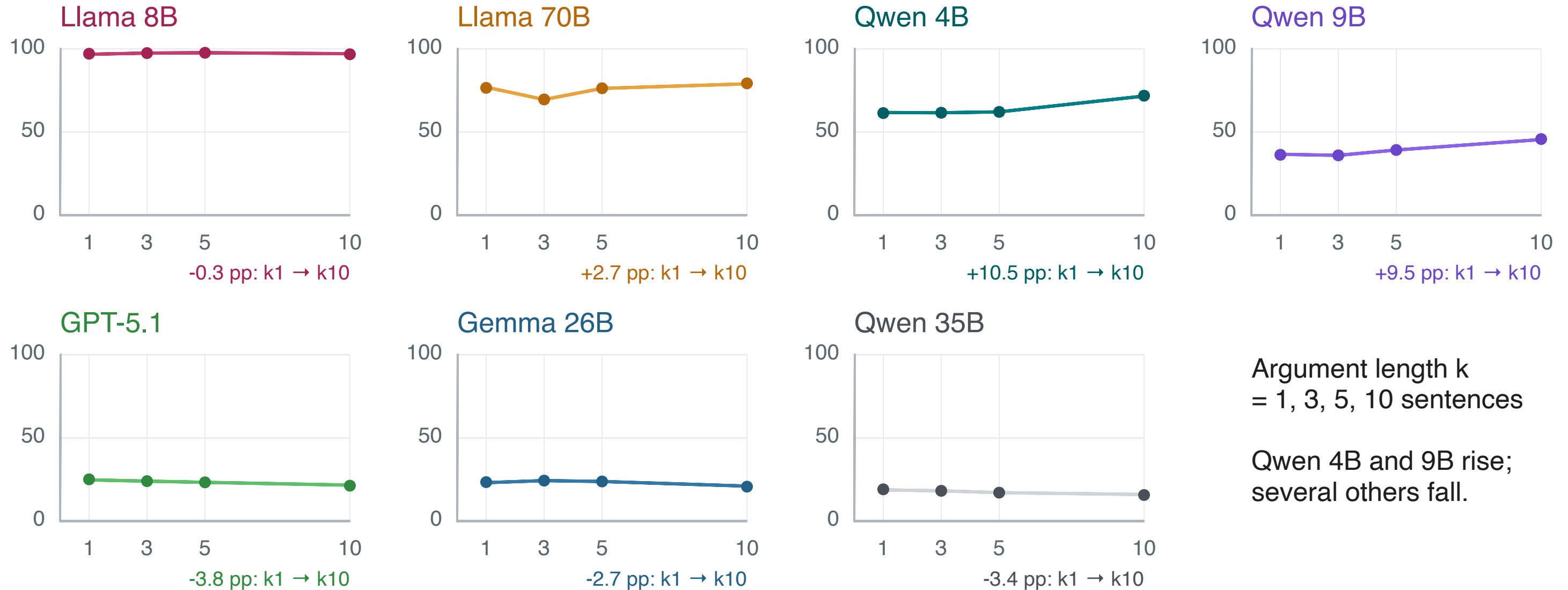
condition	argument source	presentation	sentences (k)
blind	same model	anonymous	1, 3, 5, 10
self	same model	“your earlier reasoning”	1, 3, 5, 10
cross	different model	anonymous	10 only

“Self-generated” identifies the source. “Self-attributed” describes the prompt.

Blind vs self changes attribution while keeping the underlying argument fixed.

Longer arguments push models in different directions

Seven small multiples; identical 0–100% axes; whiskers show reported 95% CIs



Pooling all models hides opposing trends; downward trends were not significant.

Eligibility and uncertainty shape the comparison

model	coverage
Llama-3.1-8B	59%
Llama-3.3-70B	80%
Qwen3.5-4B	78%
Qwen3.5-9B	81%
GPT-5.1	89%
Gemma-4-26B	87%
Qwen3.5-35B	83%

Uncertainty reporting

95% cluster-bootstrap intervals
2,000 bootstrap replicates
Cluster on the original MMLU question

Repeated options and conditions share a question.

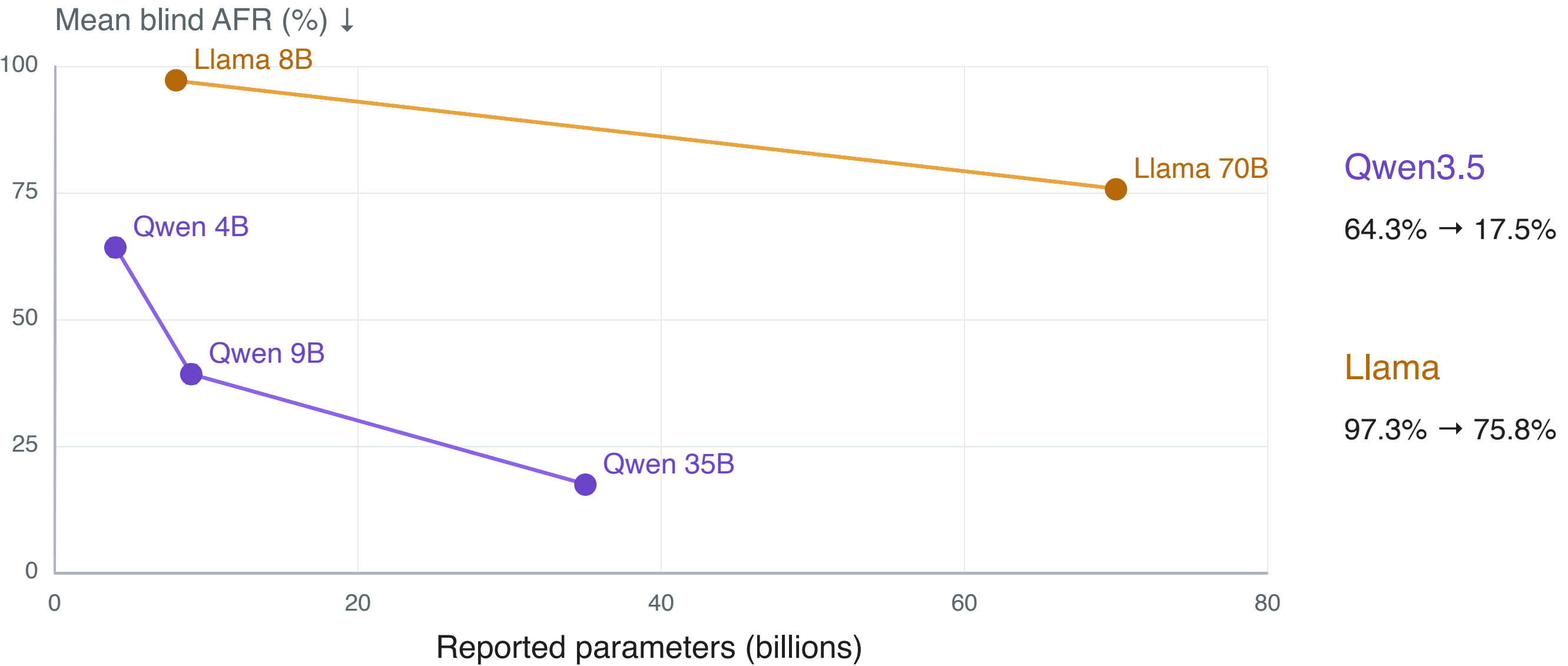
Reading the intervals

Table subscripts are CI half-widths in pp.
Coverage is eligibility, not baseline accuracy.

Models can be compared on different eligible subsets; that limits ranking claims.

Size tracks stability within Qwen, but not across families

Mean blind AFR (%) versus reported model size; Qwen and Llama family members



Total parameter labels differ from active compute; these are not controlled scaling runs.

Language differs between held and flipped answers

Descriptive associations from lexical features, not causal explanations

After the challenge

Held answers: more resistance phrases

Flipped answers: about 6x as many capitulation markers

Held responses are longer:
about 1,800 vs 1,150 characters.

Before the challenge

Later flips associate with:

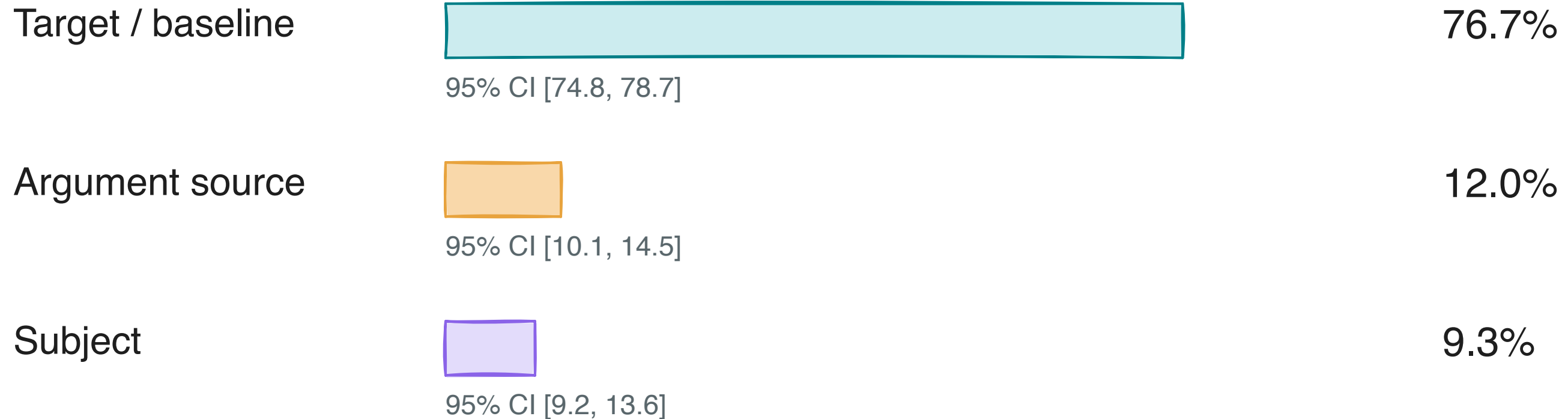
More baseline hedging
Longer baseline responses

Higher confidence in the wrong argument
does not straightforwardly predict flips.

Manually curated lexicons; case-insensitive substring matching.

Response markers partly describe the outcome; they do not establish why it happened.

Target identity dominates the reported variance



Reported decomposition across target, source and subject triples.

Source identity contributes, but target identity dominates this analysis.

Source details and reported inconsistencies

Reported numbers are preserved. These discrepancies qualify particular summaries.

detail	source evidence	reading note
Length means	Table 2 includes 47.3% at $k = 3$.	Prose range 48.4–50.2 omits it.
Subject categories	Table 5 lists 8 STEM subjects in the lowest 10.	Prose says 9 of 10.
Producer shares	Table 7 shares sum to 96.1%.	Do not normalize or invent the missing share.
Cross-matrix range	Figure 4 has 57–94% for the Llama-70B target.	Prose claims column ranges of at most 10 pp.

These checks qualify particular summaries; they do not erase the observed answer flips.